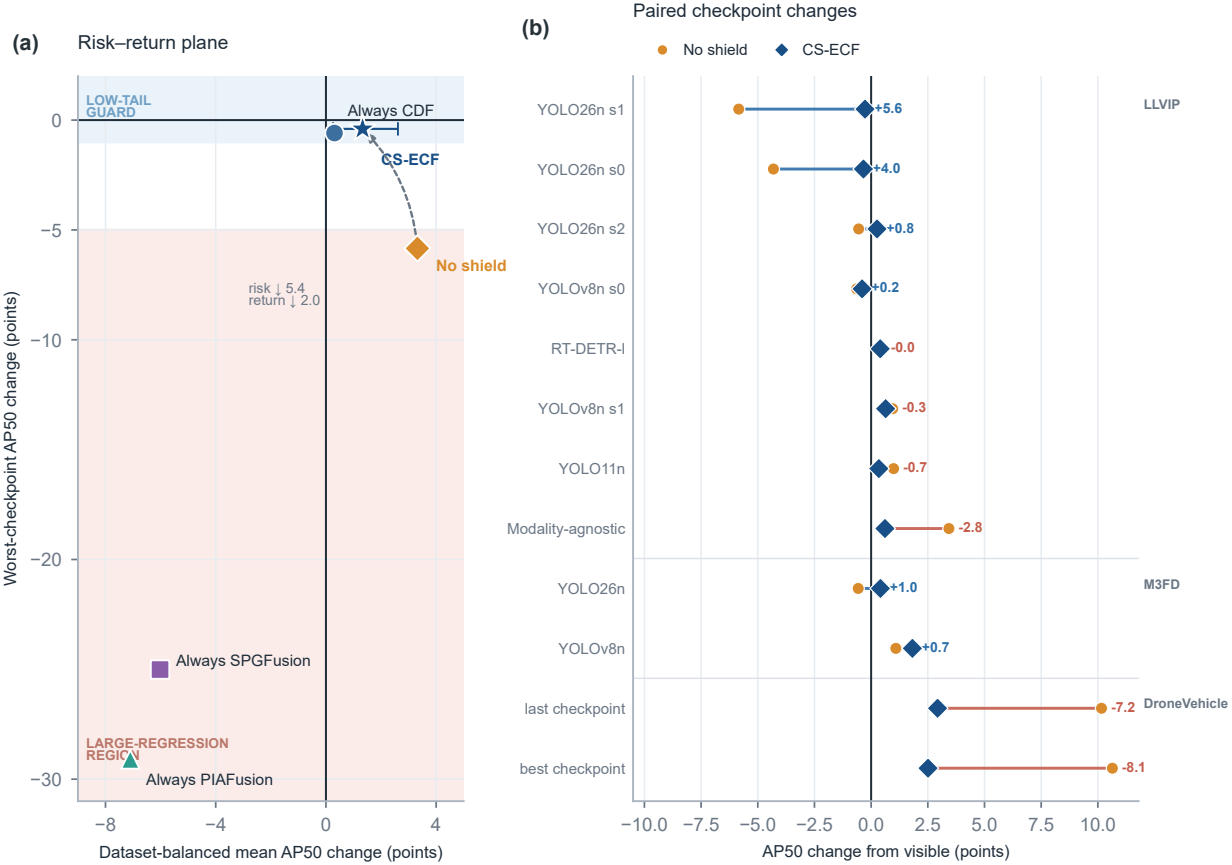

(a)
Risk–return plane
LOW-TAIL GUARD
Always CDF
CS-ECF
No shield
risk ↓ 5.4
return ↓ 2.0
Always SPGFusion
LARGE-REGRESSION REGION
Always PIAFusion
Worst-checkpoint AP50 change (points)
Dataset-balanced mean AP50 change (points)
0
−5
−10
−20
−30
−8
−4
0
4
(b)
Paired checkpoint changes
No shield
CS-ECF
YOLO26n s1
YOLO26n s0
YOLO26n s2
YOLOv8n s0
RT-DETR-l
YOLOv8n s1
YOLO11n
Modality-agnostic
YOLO26n
YOLOv8n
last checkpoint
best checkpoint
+5.6
+4.0
+0.8
+0.2
-0.0
-0.3
-0.7
-2.8
+1.0
+0.7
-7.2
-8.1
LLVIP
M3FD
DroneVehicle
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
AP50 change from visible (points)